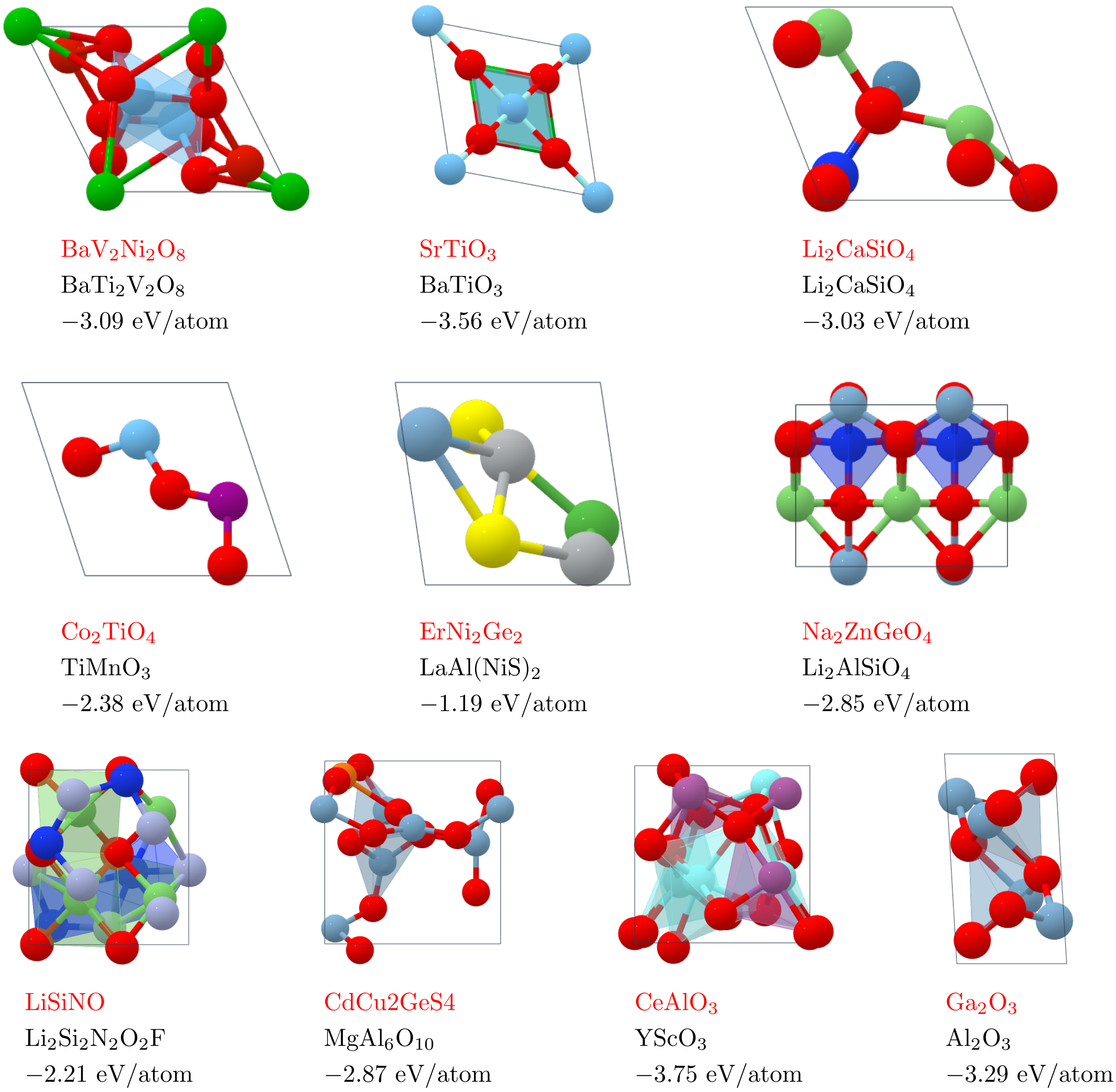

Figure D.2: Visualization of the final structures obtained by LLMatDesign for the formation energy task. These structures represent the ones with the minimum formation energy per atom from the first run of all 10 starting materials. The chemical formulae in red represent the starting materials, followed by the formulae of the structures with the lowest formation energies and their corresponding formation energy per atom values. GPT-4o with history is utilized as the LLM engine.

E DFT Calculations

Table E.1: DFT results for lowest-energy structures obtained from the formation energy task, averaged across all starting materials and runs.

	GPT-4o with history	Random
Formation energy per atom (eV/atom)	-2.31	-1.51
Job success rate (%)	73.3	40.0